Such situations arise when a new product is introduced in the market or a new plant is set up. The number of different decision criteria available under the condition of uncertainty is given below.

10.3.3 Maximin and Minimax Strategy

Maximin criteria

This criterion is the decision to take the course of action which maximizes the minimum possible pay-off. Since this decision criterion locates the alternative strategy that has the least possible loss, it is also known as a pessimistic decision criterion. The working method is:

(i) Determine the lowest outcome for each alternative.

(ii) Choose the alternative associated with the maximum of these.

Minimax criteria

This criterion is the decision to take the course of action which minimizes the maximum possible pay-off. Since this decision criterion locates the alternative strategy that has the greatest possible gain. The working method is:

(iii) Determine the highest outcome for each alternative.

(iv) Choose the alternative associated with the minimum of these.

Example 10.10

Consider the following pay-off (profit) matrix Action States

Action	States			
	(S_1)	(S_2)	(S_3)	(S_4)
A_1	5	10	18	25
A_2	8	7	8	23
A_3	21	18	12	21
A_4	30	22	19	15

Determine best action using maximin principle.

Solution:

Action	States				Minimum
	(S_1)	(S_2)	(S_3)	(S_4)	
A_1	5	10	18	25	5
A_2	8	7	8	23	7
A_3	21	18	12	21	12
A_4	30	22	19	15	15

Max (5,7,12,15)=15 ∴ Action A4 is the best.

Example 10.11

A business man has three alternatives open to him each of which can be followed by any of the four possible events. The conditional pay offs for each action - event combination are given below:

Alternative	Pay–offs (Conditional events)			
	A	*B*	*C*	*D*
X	8	0	–10	6
Y	–4	12	18	–2
Z	14	6	0	8

Determine which alternative should the businessman choose, if he adopts the maximin principle.

Solution:

Alternative	Pay – offs (Conditional events)				Minimum pay off
	A	*B*	*C*	*D*	
X	8	0	–10	6	–10
Y	–4	12	18	–2	–4
Z	14	6	0	8	**0**

Max (–10,–4, 0) = 0. Since the maximum payoff is 0, the alternative Z is selected by the businessman.

Example 10.12

Consider the following pay-off matrix.

Alternative	Pay – offs (Conditional events)			
	A_1	A_2	A_3	A_4
E_1	7	12	20	27
E_2	10	9	10	25
E_3	23	20	14	23
E_4	32	24	21	17

XII Std - Business Maths & Stat EM Chapter 10.indd 259 2/29/2020 11:45:21 AM